

# Machine Learning Models Cheat Sheet

## Linear Regression

Type: Regression

Used for predicting continuous values.

Example: Predict house prices based on area.

Tools: `sklearn.linear_model.LinearRegression`

## Logistic Regression

Type: Binary Classification

Used for classification problems with 2 outcomes.

Example: Email spam detection.

Tools: `sklearn.linear_model.LogisticRegression`

## Decision Tree

Type: Classification/Regression

Uses a tree of if-else conditions.

Example: Predict loan approval.

Tools: `sklearn.tree.DecisionTreeClassifier`

## Random Forest

Type: Classification/Regression

Ensemble of decision trees for better accuracy.

Example: Predict customer churn.

Tools: `sklearn.ensemble.RandomForestClassifier`

## Support Vector Machine (SVM)

Type: Classification

Finds optimal margin between classes.

Example: Face recognition.

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Tools: `sklearn.svm.SVC`

## **K-Nearest Neighbors (KNN)**

Type: Classification/Regression

Predicts based on nearest data points.

Example: Recommender systems.

Tools: `sklearn.neighbors.KNeighborsClassifier`

## **Naive Bayes**

Type: Text Classification

Applies Bayes' theorem with independence assumption.

Example: Sentiment analysis.

Tools: `sklearn.naive_bayes.MultinomialNB`

## **XGBoost**

Type: Classification/Regression

Boosted trees for performance.

Example: Kaggle competitions.

Tools: `xgboost.XGBClassifier`

## **K-Means Clustering**

Type: Clustering (Unsupervised)

Groups similar data points.

Example: Customer segmentation.

Tools: `sklearn.cluster.KMeans`

## **PCA (Principal Component Analysis)**

Type: Dimensionality Reduction Reduces

features while preserving variance.

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Example: Visualizing high-dimensional data.

Tools: sklearn.decomposition.PCA

## Q-Learning (Reinforcement Learning)

Type: Sequential Decision Making

Learns from rewards and penalties.

Example: Game AI, robot navigation.

Tools: custom or TensorFlow/PyTorch

| Problem Type               | Example Use Case                   | Recommended Models                        |
|----------------------------|------------------------------------|-------------------------------------------|
| Regression                 | Predict house prices               | Linear Regression, Random Forest, XGBoost |
| Binary Classification      | Spam detection, loan approval      | Logistic Regression, SVM, Random Forest   |
| Multi-class Classification | Digit recognition (0–9)            | SVM, KNN, Decision Tree, XGBoost          |
| Clustering                 | Customer segmentation              | K-Means, DBSCAN, Hierarchical Clustering  |
| Dimensionality Reduction   | Visualizing high-dimensional data  | PCA, t-SNE                                |
| Recommendation             | Movie or product recommendations   | KNN, Matrix Factorization, Deep Learning  |
| Anomaly Detection          | Fraud detection, outlier detection | Isolation Forest, One-Class SVM, DBSCAN   |
| Sequential Decision Making | Self-driving cars, game AI         | Q-Learning, DQN, Policy Gradient          |
| Text Classification        | Sentiment analysis, spam detection | Naive Bayes, Logistic Regression, SVM     |
| Image Classification       | Cat vs Dog classifier              | CNNs (Deep Learning)                      |